

# HRIJSHAM C SUNIL

**Cybersecurity Analyst | SOC Operations & Threat Detection | Penetration Testing | OT/ICS Security**

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## PROFESSIONAL SUMMARY

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Cybersecurity Engineer with hands-on experience in SOC operations, incident response, penetration testing, and OT/ICS security, combining a B.Tech in Electrical and Electronics Engineering with an Advanced Diploma in Cyber Defense and CEH certification. At NIT Calicut, executed False Data Injection (FDI) attacks on Modbus and false command injection attacks on IEC 61850 GOOSE messaging to evaluate cyber resiliency of power transmission and distribution systems, applying MITRE ATT&CK-aligned threat modeling alongside blockchain- and edge-computing-based mitigation. Performed security monitoring, SIEM-based log analysis, and incident escalation as an Information Security Systems intern at Geojit. Delivered ethical hacking, penetration testing, and incident response training to 300+ students and professionals, including 50+ training sessions and CTF events. Proficient in Splunk, Suricata, Nessus, Wireshark, Metasploit, Burp Suite, and Nmap, with working knowledge of NIST CSF, ISO 27001, and MITRE ATT&CK, plus industrial protocols (Modbus, IEC 61850), SCADA, and Microsoft Defender. Seeking to apply this cross-disciplinary security and engineering background in a SOC Analyst, Cybersecurity Analyst, Penetration Tester, or OT/ICS Security role.

## CORE COMPETENCIES

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Security Operations (SOC) • Incident Response • Threat Detection & Monitoring • SIEM & SOAR Log Analysis • Penetration Testing • Ethical Hacking • Vulnerability Assessment • Network Security • OT/ICS Security • Red Team & Blue Team Exercises • Security Awareness Training • Antivirus/SOC Reporting • Security Audit Support • NIST CSF • ISO 27001 • MITRE ATT&CK • OSINT • OWASP

## TECHNICAL SKILLS

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**SIEM & Security Monitoring:** Splunk, Splunk Enterprise, Splunk Universal Forwarder, Seceon, Microsoft Defender, CloudSek, EDR & XDR

**Network Security & Traffic Analysis:** Wireshark, Suricata, Network Security, Intrusion Detection

**Vulnerability Management:** Nessus, Vulnerability Assessment, Vulnerability Scanning, Audit Support

**Offensive Security / Penetration Testing:** Metasploit, Burp Suite, Nmap, Kali Linux, P4wnP1 A.L.O.A.

**OT/ICS & Industrial Security:** Modbus, IEC 61850 (GOOSE), SCADA, OPAL-RT, Edge Computing, Blockchain-Based Cyber Resiliency, False Data Injection (FDI) Attacks, GOOSE Message/Command Injection Attacks

**Security Frameworks & Standards:** NIST Cybersecurity Framework (CSF), ISO/IEC 27001, MITRE ATT&CK

**Programming & Scripting:** Python, MATLAB

**Platforms & Tools:** Docker, Antivirus/SOC Reporting Tools

**Engineering Foundations:** Circuit Design and Analysis, Control Systems, Power Electronics

## PROFESSIONAL EXPERIENCE

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### Cybersecurity Engineer

*Crimson Energy Experts Pvt Ltd, Kochi, Kerala | August 2026 – Present*

- Cybersecurity Engineer with hands-on experience securing Operational Technology (OT), Industrial Control Systems (ICS), and SCADA environments across critical infrastructure and manufacturing operations.
- Executed False Data Injection (FDI) attacks against the Modbus protocol and false command injection attacks against IEC 61850 GOOSE messaging to evaluate the cyber resiliency of integrated transmission and distribution systems against OT/ICS-targeted threats.

### Project Associate

*National Institute of Technology (NIT) Calicut, Kerala | December 2025 – March 2026*

- Contributed to the research project “Edge Computing Assisted Blockchain-Based Cyber Resiliency Enhancement of Integrated Transmission and Distribution Systems,” conducting protocol-level attack simulations on power grid communication systems.
- Executed False Data Injection (FDI) attacks against the Modbus protocol and false command injection attacks against IEC 61850 GOOSE messaging to evaluate the cyber resiliency of integrated transmission and distribution systems against OT/ICS-targeted threats.
- Applied MITRE ATT&CK-aligned threat modeling and blockchain/edge-computing-based mitigation concepts to strengthen resilience of critical power infrastructure against the simulated attacks.

### **Information Security System (ISS) – Intern**

*Geojit, Edapally, Kerala | October 2025 – December 2025*

- Performed system security checks across endpoints to prevent data exfiltration and ensure proper data sanitization, strengthening the organization’s data protection posture.
- Monitored security events and escalated incidents through SIEM tooling, supporting timely detection and response within SOC workflows.
- Supported internal security audits and prepared antivirus/SOC status reports for the Information Security team.

### **Cybersecurity Researcher & Trainer**

*RedTeam Hacker Academy, Kottakkal, Kerala | June 2024 – June 2025*

- Delivered hands-on training to 300+ students and professionals across core cybersecurity domains, including Ethical Hacking, Penetration Testing, and Incident Response.
- Designed and implemented a complete ethical hacking curriculum, translating offensive security tools (Metasploit, Burp Suite, Nmap, Wireshark) into structured, outcome-based lessons.
- Conducted 50+ training sessions and Capture the Flag (CTF) events, building practical SOC and penetration testing skills among learners new to cybersecurity.

## **CYBERSECURITY PROJECTS**

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### **Splunk SIEM Lab Deployment | Tools: Splunk Enterprise, Splunk Universal Forwarder, Kali Linux**

- Deployed and configured Splunk Enterprise and Universal Forwarder, then simulated attacks from Kali Linux to test real-time alerting and detection of suspicious behavior — reinforcing SOC monitoring and SIEM administration skills.

### **Wireless Attack Toolkit | Tools: Microcontrollers, Wi-Fi/Bluetooth jamming hardware**

- Designed and built a Wi-Fi/Bluetooth jammer and deauther to demonstrate wireless network vulnerabilities, identifying exploitation paths and proposing mitigation approaches.

### **P4wnP1 A.L.O.A. Framework | Tools: Raspberry Pi Zero W, P4wnP1 A.L.O.A. framework**

- Converted a Raspberry Pi Zero W into a covert USB attack platform for authorized physical penetration testing and payload delivery, enabling hands-on red team exercises.

### **IoT-Based Smart Compost Bin (B.Tech Final Year Project) | Tools: Embedded systems, IoT sensors, solar power**

- Integrated sensors and solar power into an embedded system for automated waste monitoring, demonstrating embedded systems and IoT integration skills relevant to IoT security.

### **Automatic Braking System | Tools: Sensors, embedded control systems**

- Designed a real-time obstacle detection and automatic braking system, demonstrating real-time control system design and safety-critical systems thinking.

## **CERTIFICATIONS**

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- Certified Ethical Hacker (CEH)
- CISA ICS CYBERSECURITY 300

## **EDUCATION**

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### **Advanced Diploma in Cyber Defense**

*RedTeam Hacker Academy | April 2023 – January 2024*

- Core Training: Ethical Hacking, Penetration Testing, Network Security
- Tools: Splunk, Nmap, Metasploit, Wireshark, Burp Suite, Suricata

### **Bachelor of Technology (B.Tech) – Electrical and Electronics Engineering**

*Calicut University Institute of Engineering and Technology (CUIET) | August 2018 – November 2022*